

L13

For real numbers w and z , $\frac{\frac{1}{2w} + \frac{1}{z}}{\frac{1}{w} - \frac{1}{z}} = 2022$. What is $\frac{w-z}{2w+z}$? (Adapted From 2014 AMC 10B Problem, Question #9)

A. -4044

B. -2022

C. $-\frac{1}{4044}$

D. $\frac{1}{4044}$

E. $\frac{1}{2022}$

L13 HW

3

$$(2) \frac{x-2}{x+3} - \frac{x+3}{x-2} = \underline{\hspace{2cm}}$$

9 Simplify $\left(1 + \frac{1}{x-1}\right) \div \frac{1}{x^2-1} + (x-3)$, and then what is the value when $x = \sqrt{5}$?

A. 2

B. 8

C. $2 + 2\sqrt{5}$

D. $2 - 2\sqrt{5}$

10 Simplify $\left(a + \frac{1+a}{a-1}\right) \div (a^2 + 1)$, and then find the value when $a = \sqrt{3} + 1$.

The value is _____.

L13 Extention

2 Simplify:

(1) $\frac{2x^2 + 1}{3x - 4} - \frac{x^2}{4 - 3x} = \underline{\hspace{2cm}}$

A. $\frac{3x^2 + 1}{4 - 3x}$

B. $\frac{x^2 + 1}{4 - 3x}$

C. $\frac{x^2 + 1}{3x - 4}$

D. $\frac{3x^2 + 1}{3x - 4}$

3 Simplify: $\frac{1}{x-2} - \frac{1}{y-2} = \underline{\hspace{2cm}}$.

A. $\frac{-x+y}{(x-2)(y-2)}$

B. $\frac{x-y}{(x-2)(y-2)}$

C. $\frac{-x-y}{(x-2)(y-2)}$

D. $\frac{x+y}{(x-2)(y-2)}$

7 If $a + \frac{1}{b} = 1$, $b + \frac{1}{c} = 1$, then $c + \frac{1}{a} = \underline{\hspace{2cm}}$.

A. 1

B. $\frac{1}{b-1}$

C. $\frac{1}{a(b-1)}$

D. $\frac{1}{a-1}$

10 Simplify $\left(a + \frac{1+a}{2a-1}\right) \div (2a^2 + 1)$, and then find the value when $a = \sqrt{2} + \frac{1}{2}$.

The value is $\underline{\hspace{2cm}}$.

6 Solve the following rational equation: $\frac{1}{1-x} - 1 = \frac{2}{x-1}$

$x = \underline{\hspace{2cm}}$.

L14

3 Solve the rational equation $\frac{x+1}{x-1} = 1 - \frac{4}{1-x^2}$.

A. $x = 1$

B. $x = -1$

C. $x = \pm 1$

D. No solution

L14 HW

3

(2) $\frac{x-2}{x+3} - \frac{3}{x-3} = 1$

$x = \underline{\hspace{2cm}}$.

5

(1) $\frac{3}{2x-4} - \frac{x}{x-2} = \frac{1}{2}$

$x = \underline{\hspace{2cm}}$.

5

$$(3) \frac{2}{x+3} + \frac{3}{2} = \frac{7}{2x+6}$$

$$x = \underline{\hspace{2cm}} .$$

6 Solve the following rational equations:

$$(1) \frac{3}{x^2-9} + \frac{x}{x-3} = 1$$

$$x = \underline{\hspace{2cm}} .$$

6

$$(2) \frac{x+1}{4x^2-1} = \frac{3}{2x+1} - \frac{4}{4x-2}$$

$$x = \underline{\hspace{2cm}} .$$

- 3 If A and B are two constants and $\frac{3x}{x^2 - 7x + 12} = \frac{A}{x - 3} + \frac{B}{x - 4}$ ($x \neq 3$ and $x \neq 4$), then $A = \underline{\hspace{1cm}}$ and $B = \underline{\hspace{1cm}}$.

L14 Extention

- 8 Solve the rational equation $\frac{x - 1}{x + 1} = 1 - \frac{4}{1 - x^2}$.

A. $x = 1$

B. $x = -1$

C. $x = \pm 1$

D. No solution

7 Solve the following rational equation: $\frac{x^2}{x^2 + 4x + 4} - \frac{1}{x + 2} = 1$

$x = \underline{\hspace{2cm}}$.

- 10 To protect arable land, a certain area needs to convert 1500 acres of forest back to arable land. It is known that the number of acres Team *A* can convert back to arable land per day is 1.2 times that of Team *B*. If each team completes the task alone, Team *A* would take 5 days less than Team *B*. How many acres can Teams *A* and *B* complete per day?

L15

2 Which of the following equations is always a quadratic equation?

A. $ax^2 + x + 1 = 0$

B. $x^2 = 0$

C. $\left(\frac{1}{x}\right)^2 + \frac{2}{x} + 1 = 0$

D. $x(x - 1) = x^2$

L15 HW

6 There are two values of a for which the equation $x^2 + ax + 6x + 16 = 0$ has only one solution for x . The sum of these values of a is _____ (Adapted from 2005 AMC 12A Problems, Question #9)

10 Let d and e denote the solutions of $3x^2 + 7x - 6 = 0$. The value of $(d + 4)(e + 4)$ is _____. (Adapted from 2003 AMC 10A Problems, Question #5)

L15 Extention

8 Solve the following quadratic equations:

(1) $2x^2 - 3x - 2 = 0$

$x = \underline{\hspace{2cm}}$ and $x = \underline{\hspace{2cm}}$.

9 Let d and e denote the solutions of $4x^2 + 4x - 3 = 0$. The value of $(d + \frac{1}{2})(e + \frac{1}{2})$ is $\underline{\hspace{2cm}}$.

L16

Math Exploration 4.1

A quadratic equation $ax^2 - 2ax + b = 0$ has two real solutions. The average of these two solutions is _____. (2008 AMC 10B Problems, Question #9)

L16 HW

2 Solve the following quadratic equations by completing the square:

(1) $3x^2 - 5x - 2 = 0$, $x = \underline{\hspace{1cm}}$ and $x = \underline{\hspace{1cm}}$.

$\frac{5 \pm \sqrt{49}}{6}$

(2) $5x^2 = 6x + 8$, $x = \underline{\hspace{1cm}}$ and $x = \underline{\hspace{1cm}}$.

$\frac{3 \pm \sqrt{41}}{5}$

L16 Extention

7 Solve the following quadratic equations by completing the square:

(1) $x^2 - 8x + 10 = 0$

$x = \underline{\hspace{2cm}}$ and $x = \underline{\hspace{2cm}}$.

(2) $x^2 - 2x - 3 = 0$

$x = \underline{\hspace{2cm}}$ and $x = \underline{\hspace{2cm}}$.

2 Solve the following quadratic equations by completing the square:

(1) $2x^2 + 5x + 1 = 0$, $x = \underline{\hspace{2cm}}$ and $x = \underline{\hspace{2cm}}$.

(2) $4x^2 = 6x + 3$, $x = \underline{\hspace{2cm}}$ and $x = \underline{\hspace{2cm}}$.

(3) $2x^2 - 5x - 3 = 0$, $x = \underline{\hspace{1cm}}$ and $x = \underline{\hspace{1cm}}$

- 5 Apply the Quadratic Formula to solve the following problem and verify your answer by Completing the Square.

There is a positive integer n such that $(n + 3)! + 3(n + 2)! = (n + 1)! \cdot 21$. The value of n is $\underline{\hspace{1cm}}$. (Adapted from 2019 AMC 10B Problems, Question #6)

- 7 If the quadratic equation $(k + 1)x^2 + 3x + 1 = 0$ in terms of x has two different solutions, the range of k is $\underline{\hspace{1cm}}$.

- 9 If the quadratic equation $x^2 + 2x - 2m = 0$ in terms of x has two different solutions, the range of m is _____.

- 5 Solve the following quadratic equations by the quadratic formula:

(1) $x^2 - 6x - 2 = 0$, $x = \underline{\hspace{2cm}}$ and $x = \underline{\hspace{2cm}}$.

- 10 The sum of the reciprocals of the roots of the equation $2020x^2 + 2019x + 2018 = 0$ is _____. (Adapted from 2003 AMC 10A Problems, Question #18)

- 7 Three families, each with 3 family members, sit in a row of 9 seats. If family members sit together, there are _____ different possible arrangements in total.
- A. $3 \times 3!$ B. $3 \times (3!)^3$ C. $(3!)^4$ D. $9!$
- 2 A dessert chef prepares the dessert for every day of a week starting with Sunday. The dessert each day is either cake, pie, ice cream, or pudding. The same dessert may not be served two days in a row. There must be cake on Friday because of a birthday. How many different dessert menus for the week are possible? (2012 AMC 10B Problem, Question #11)
- A. 729 B. 972 C. 1024 D. 2187 E. 2304

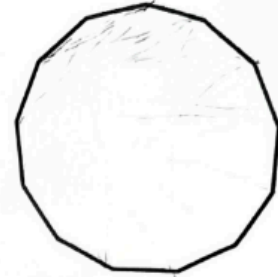
- 4 How many even three-digit integers have the property that their digits, read left to right, are in strictly increasing order? (2006 AMC 12B Problem, Question #9)

A. 21 B. 34 C. 51 D. 72 E. 150

- 5 The number 2013 has the property that its units digit is the sum of its other digits, that is $2 + 0 + 1 = 3$. How many integers are less than 2013 but greater than 1000 share this property?

A. 33 B. 34 C. 45 D. 46 E. 58

- 7 There are a total of _____ triangles with different shapes whose vertices are the vertices of a regular tridecagon (13-sided polygon). (Congruent triangles are considered the same shape)



A. 14 B. 13 C. 12 D. 11 E. 10

- 10 In how many ways can a party of 4 men and 4 women be seated at a circular table so that no two women are adjacent? (If one arrangements can be obtained by rotating another arrangement, they are considered the same arrangement).

A. 40320 B. 720 C. 576 D. 144 E. 36

L17 Extention

- 3 There is an advancing 8-person queue where each person's height is different and they are arranged in a ascending order from the shortest to the tallest. Now they need to make a 2-by-4 formation, with 2 rows and 4 columns. If people in the same row from left to right are still arranged in a ascending order from the shortest to the tallest, and it is also required that for each pair of adjacent people in the same column, the person in front should be shorter than the one behind. Then, there are _____ different ways to arrange the 8 people.

- 4 Consider integers 0, 1, 2, 3, 4, 5:

(3) We can form _____ four-digit numbers that are greater than 3000 but less than 5421 without repeating digits.

7 Three-digit numbers satisfying $a + c = 2b$ amount to () .

A. 16

B. 36

C. 45

D. 49

2 Five friends a , b , c , d , and e gathered in the park and greeted each other with handshakes. It is known that a shook hands 4 times, b shook hands 1 times, c shook hands 3 times, and d shook hands 2 times. So far, e shook hands _____ times.

8 In a class of 50 pupils, 32 pupils like to eat fish, 40 pupils like to eat chicken and 46 pupils like to eat vegetables. What is the least number of pupils who like all three types of food?

L18 HW

- 7 Aaron, Brian, and Claire are in an ice cream shop. There are 8 different flavors, and they would like to order one scoop of each flavor. If two of them get 3 scoops, and the other gets 2 scoops, how many different ways are there to give them the ice cream?

A. 1680 B. 3360 C. 18816 D. 2352 E. 336

- 6 Put 10 identical balls into 7 bins, with at least one ball in each bin. How many different ways to distribute the balls are there?

A. 72 B. 75 C. 80 D. 84 E. 90

7 There are 8 parking spaces and 4 different cars. How many different possible ways to park these cars so that exactly 3 of the empty parking spaces are next to each other?

A. 1680

B. 840

C. 480

D. 360

E. 240

9 If 3 boys, including Jack and Andy, and 3 girls line up in a row, which of the following statements is incorrect?

A. There are 720 different arrangements.

B. The number of arrangements where Jack is positioned at either end is 240.

C The total number of arrangements where Jack and Andy are adjacent is 120.

D The total number of arrangements where boys and girls alternate is 72.

- 10 Choose 5 distinct letters from the word 'equation' and arrange them in a row. How many different arrangements contain the substring 'qu', where 'qu' must remain adjacent and in order?

A. 120

B. 480

C. 720

D. 840

L18 Extention

- 5 There are 10 people in a party and each shakes hand with everyone else only once. How many handshakes take place during the party?

- 7 There are six people, A, B, C, D, E , and F , who are divided into three groups: red, blue, and green. Each group consists of two people to play a game. There are a total of _____ different ways to divide them.

- 9 DeepSeek software is a deeply exploratory artificial intelligence software developed in China. Now rearrange the letters of the word DeepSeek. The number of different arrangements where the letters *e* are not adjacent is _____ .

A. 24 B. 120 C. 576 D. 2880

- 10 *A, B, C, D, E, F* form a line, satisfying the conditions that *A, B* are adjacent, *C, D* are not adjacent, and *E* does not stand at the two ends. How many different arrangements are there?

A. 48 B. 96 C. 72 D. 144 E. 288

- 1 How many ways can 8 different books be arranged on a shelf that can hold 3 books?

- 8 5 boys and 4 girls are going to attend a stage show. There are exactly 5 different male characters and 4 different female characters in the show. There are _____ ways to assign the roles to them.

- 9 A team has 5 boys, 4 girls. Now we need to pick 3 people from this team to do the cleaning.

(1) If we are to pick at least 2 boys, there are _____ ways of picking.

(2) If we are to pick at least 1 girls, there are _____ ways of picking.